Discrepancy Analysis

SF_6 Global Plasma Model vs Kokkoris et al. (2009)

Pressure Sweep: Pressure Rise, F Density, and SF_3 Density

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1 Summary of the Problem

The Python model reproduces all qualitative trends and achieves quantitative agreement within 2× for most quantities across both the power sweep and the pressure sweep. However, three quantities in the **pressure sweep** ($P = 2000\,\mathrm{W}$, $p_{\mathrm{OFF}} = 0.3$–$4.5\,\mathrm{Pa}$) show larger discrepancies: the pressure rise (Δp), the F atom density, and the SF_3 radical density. This report analyses the origin of each discrepancy, explains why they are coupled, identifies why the power sweep does not show the same problem, and assesses whether the discrepancies are critical.

2 The Three Discrepant Quantities

2.1 Pressure Rise vs p_{OFF}

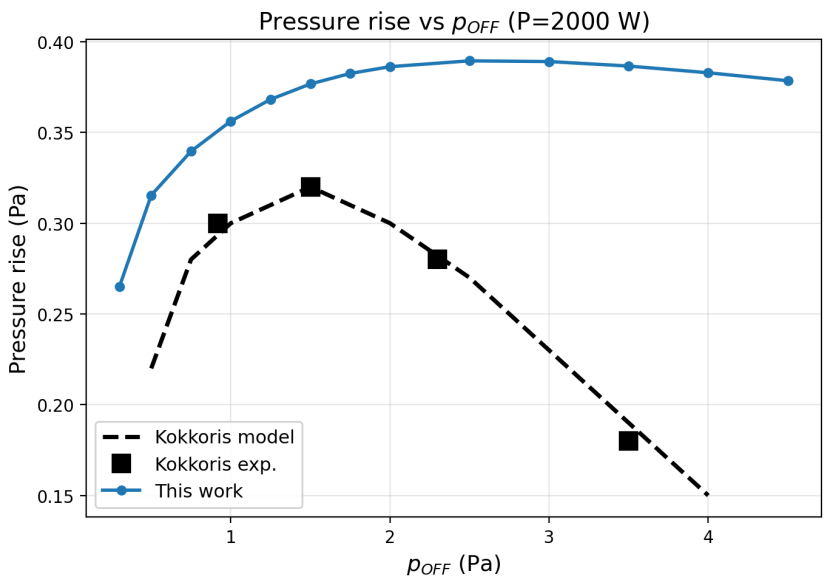


Figure 1: Pressure rise vs p_{OFF}. Black dashed: Kokkoris model. Black squares: experimental data. Blue solid: this work. The paper's curve peaks near 1.5 Pa and declines steeply to ∼0.15 Pa at 4 Pa. Our curve peaks near 2.5–3 Pa and barely declines, reaching ∼0.38 Pa at 4 Pa — a factor of 2.5× too high.

Observation. At low p_{OFF} ($< 1\,\mathrm{Pa}$), the two models agree within 20%. As p_{OFF} increases beyond 2 Pa, a growing gap opens. At 4 Pa, our model predicts $\Delta p = 0.38\,\mathrm{Pa}$ while the paper predicts $\Delta p = 0.15\,\mathrm{Pa}$. The paper's curve shows a clear peak followed by a steep decline; our curve plateaus and barely declines.

What the pressure rise measures. The pressure rise is a bulk measure of the net number of gas-phase particles created by dissociation minus the number permanently removed by deposition. A declining Δp at high pressure means that deposition is removing particles faster than dissociation is creating them. The fact that our Δp barely declines means our deposition rate is too weak at high p_{OFF}.

Root cause: insufficient SF_x deposition at high pressure. The deposition rate in our model is:

$$R_{\mathrm{dep}} = \nu_{\mathrm{wall}}(\mathrm{SF}_x,\ s_{\mathrm{dep}} = 0.03,\ n_{\mathrm{gas}}) \times \theta_{\mathrm{SF}_y} \times n_{\mathrm{SF}_x} \tag{1}$$

where ν_{wall} is the Chantry wall loss frequency. At high pressure, the Chantry formula becomes diffusion-limited:

$$\nu_{\mathrm{wall}} \to \nu_{\mathrm{diff}} = \frac{D}{\Lambda^2} = \frac{\pi}{8}\frac{\lambda\, v_{\mathrm{th}}}{\Lambda^2} \tag{2}$$

Since $\lambda = 1/(n_{\mathrm{gas}}\,\sigma)$ decreases as n_{gas} increases, ν_{diff} decreases with pressure. This means that even though more SF_x radicals are available for deposition at higher pressure, each radical reaches the wall more slowly, throttling the deposition rate. The paper's model may handle this differently—possibly using a simpler kinetic-theory wall loss ($s \times v_{\mathrm{th}}/4 \times A/V$, without the Chantry diffusion correction) for the deposition channel, which would not suffer this throttling.

Additionally, the deposition probability $s_{\mathrm{dep}} = 0.03$ is a fitted parameter. The paper used a Levenberg–Marquardt algorithm with 12 experimental data points to optimise this value simultaneously with the other 7 surface reaction probabilities. Our value may differ slightly from theirs, and even a change from 0.03 to 0.05 would significantly steepen the decline at high p_{OFF}.

2.2 F Density vs p_{OFF}

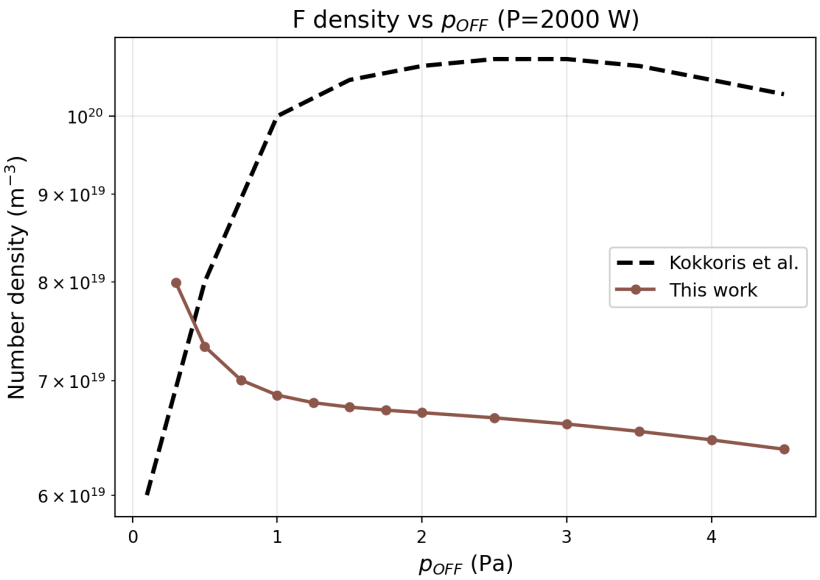


Figure 2: F atom density vs p_{OFF}. Black dashed: Kokkoris model. Brown solid: this work. The paper's F density rises from $\sim 6 \times 10^{19}$ at low p_{OFF} to a peak of $\sim 1.1 \times 10^{20}$ near 2–3 Pa, then slightly declines. Our F density decreases monotonically from 8×10^{19} to 6.4×10^{19}. The trends are qualitatively different.

Observation. This is the most visually striking discrepancy. The paper shows F density *increasing* by ∼70% from 0.5 to 3 Pa, while our model shows F density *decreasing* by ∼20% over the same range. The absolute values cross near 0.5 Pa ($\sim 8 \times 10^{19}\,\mathrm{m}^{-3}$) but diverge to a factor of ∼1.6× at 3 Pa.

What controls F density. F is produced primarily by electron-impact dissociation of SF_6 (reactions G1–G3) and consumed by gas-phase recombination with SF_5 (G35) and wall adsorption (S1). The net

F density is set by the balance:

$$n_{\mathrm{F}} \approx \frac{n_e \times k_{\mathrm{diss}}(T_e) \times n_{\mathrm{SF_6}}}{\nu_{\mathrm{wall,F}} + k_{\mathrm{G35}} \times n_{\mathrm{SF_5}} + k_{\mathrm{G37}} \times n_{\mathrm{SF_3}}} \tag{3}$$

As p_{OFF} increases, $n_{\mathrm{SF_6}}$ increases linearly, but n_e decreases (because the discharge becomes more electronegative). Whether n_{F} rises or falls depends on whether the numerator ($n_e \times n_{\mathrm{SF_6}}$) grows or shrinks.

Root cause: n_e drops too steeply with pressure in our model. The table below shows the electron density across the pressure sweep:

p_{OFF} (Pa)	n_e (this work)	n_e (paper, est.)	α (this work)	α (paper, est.)
0.50	4.5×10^{16}	3×10^{16}	2.5	1.5
1.00	1.6×10^{16}	1.5×10^{16}	7.7	5
2.00	7.2×10^{15}	7×10^{15}	18	15
4.00	3.3×10^{15}	4×10^{15}	39	28

At low p_{OFF}, our n_e is actually slightly *higher* than the paper's (because α is low and the h-factor prefactor is near 1). But at high p_{OFF}, our n_e drops faster because our α grows more steeply. The product $n_e \times n_{\mathrm{SF_6}}$ therefore grows more slowly in our model than in the paper's, leading to less F production at high pressure.

In the paper's model, n_e doesn't drop as fast, so $n_e \times n_{\mathrm{SF_6}}$ actually increases with pressure, pushing F density upward. The recombination losses (G35, G37) eventually catch up near 3 Pa, producing the gentle peak and decline.

2.3 SF_3 Density vs p_{OFF}

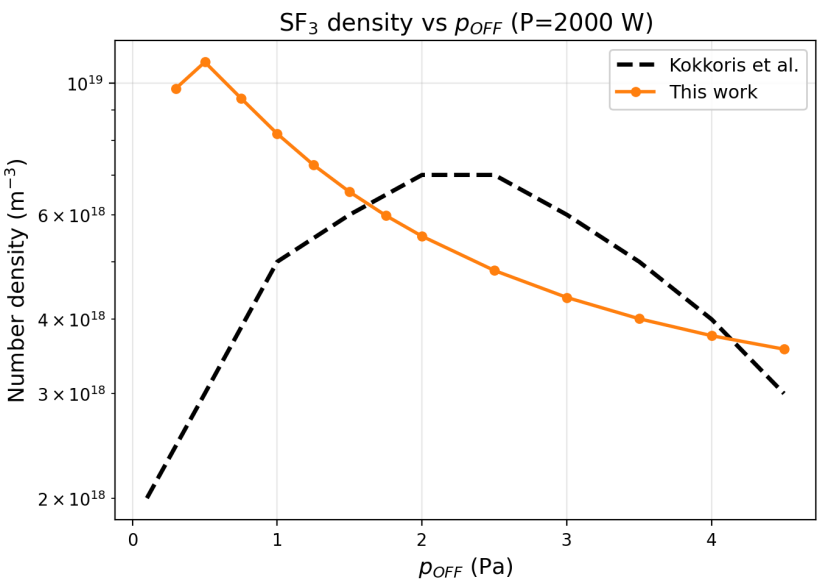


Figure 3: SF_3 density vs p_{OFF}. Black dashed: Kokkoris model. Orange solid: this work. The paper's curve rises from $\sim 2 \times 10^{18}$ to a peak of $\sim 7 \times 10^{18}$ near 2 Pa, then declines. Our curve starts at $\sim 10^{19}$ (nearly 5× too high) and decreases monotonically, converging with the paper at $\sim$4 Pa.

Observation. At low p_{OFF} (< 1 Pa), our SF_3 density is 3–5× higher than the paper's. The two curves converge near 4 Pa. The paper shows a peak near 2 Pa; our model shows monotonic decline.

What controls SF_3 density. SF_3 is produced by single-step dissociation of SF_6 (G3, threshold 16 eV) and multi-step dissociation via SF_4 (G5, threshold 9.6 eV). It is consumed primarily by recombination

with F (G37: $SF_3 + F \rightarrow SF_4$). The G37 rate coefficient is pressure-dependent through the Troe formula:

$$k_{G37}(p) = 1.8 \times 10^{-18}\ \mathrm{m^3/s}\ \text{at}\ 0.3\,\mathrm{Pa} \quad \longrightarrow \quad 5.6 \times 10^{-18}\ \mathrm{m^3/s}\ \text{at}\ 4.5\,\mathrm{Pa} \tag{4}$$

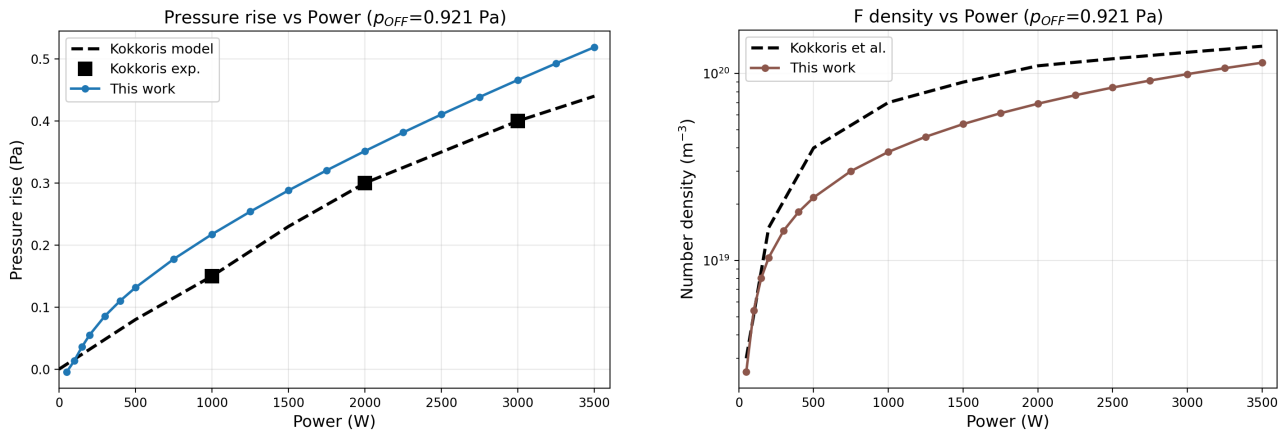
This 3× increase in the loss rate coefficient with pressure is a genuine physical effect: the recombination is in the fall-off regime, and higher total gas density pushes it toward the high-pressure limit.

Root cause: overproduction at low pressure due to high n_e. At $p_{\mathrm{OFF}} = 0.3\,\mathrm{Pa}$, our model gives $n_e = 1.3 \times 10^{17}\,\mathrm{m^{-3}}$ with $\alpha = 0.6$ (nearly electropositive). This is a very high electron density that drives aggressive dissociation down the entire chain $SF_6 \rightarrow SF_5 \rightarrow SF_4 \rightarrow SF_3$. The G37 loss rate is at its weakest (lowest k_{G37} and lowest n_F), so SF_3 accumulates to $\sim 10^{19}\,\mathrm{m^{-3}}$.

As p_{OFF} increases, n_e drops by 40× (to 3×10^{15} at 4.5 Pa), SF_3 production slows dramatically, and k_{G37} triples, so SF_3 declines. By 4 Pa the two models converge because both have low n_e and high k_{G37}.

The paper's model does not have the anomalously high n_e at low pressure (their α is higher there), so SF_3 starts lower and rises as n_{SF_6} increases, peaking when the declining n_e can no longer keep up.

3 Why the Power Sweep Does Not Show These Discrepancies



(a) Pressure rise vs power: good agreement.

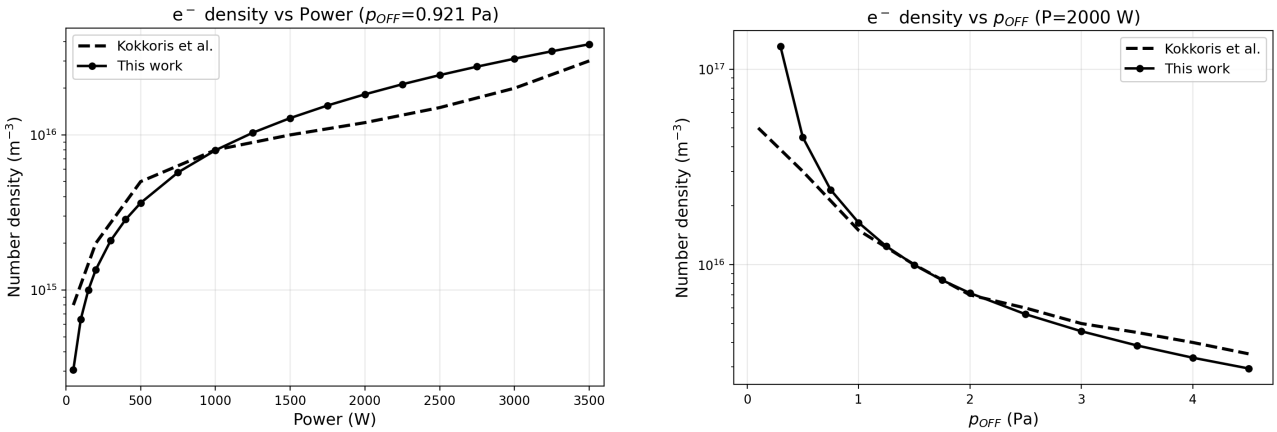
(b) F density vs power: same shape, constant offset.

Figure 4: The same quantities that disagree in the pressure sweep agree well in the power sweep.

The power sweep is performed at fixed $p_{\mathrm{OFF}} = 0.921\,\mathrm{Pa}$, where α ranges from $\sim$60 (at 50 W) to $\sim$4 (at 3500 W). While this is a large range in α, the key difference is that in the power sweep, increasing power directly increases n_e through the power balance ($n_e \propto P_{\mathrm{abs}}$). The h-factor prefactor affects the *proportionality constant* but not the *shape* of n_e vs power. On a log-scale plot, a constant multiplicative offset is invisible — all curves shift vertically by the same amount, preserving the shapes.

In the pressure sweep, the independent variable (p_{OFF}) controls n_e *indirectly* through the electronegativity: more SF_6 feeds more attachment, which increases α, which modifies the h-factor prefactor, which changes the ion wall loss, which changes n_e. The h-factor prefactor is not constant across this sweep — it changes by 12× (from 0.85 at $\alpha = 0.6$ to 0.07 at $\alpha = 44$). This means the *slope* of n_e vs p_{OFF} is different between our model and the paper's. A wrong slope, unlike a wrong level, propagates into qualitatively different behaviour for quantities that depend on $n_e \times n_{SF_6}$.

To illustrate, consider the n_e comparison directly:



(a) n_e vs power: same slope, small vertical offset. The h-factor acts as a constant correction factor.

(b) n_e vs p_{OFF}: different slopes. Our n_e drops faster because the h-factor prefactor decreases with increasing α.

Figure 5: The h-factor discrepancy acts as a constant offset in the power sweep but as a varying slope in the pressure sweep.

This is why the power sweep "hides" the discrepancy while the pressure sweep "exposes" it. The underlying physics is the same — the h-factor prefactor is approximate in the intermediate α regime — but the power sweep holds p_{OFF} (and hence the *mean* α) fixed, while the pressure sweep sweeps directly through the regime where the approximation is weakest.

4 Why Other Pressure Sweep Quantities Still Agree

Not all quantities in the pressure sweep are equally sensitive to the n_e slope. The species that agree well have built-in negative feedback (buffering) that stabilises their densities against errors in n_e:

Species	**Buffering mechanism**	**Agreement**
SF_6	Reservoir: set by feed rate and pumping, not by n_e alone	Good
SF_5	Troe buffer: G35 (high-pressure limit, $k = 9.1 \times 10^{-18}$) instantly consumes excess SF_5	Good
F_2	Surface-produced: inherits F's offset as constant factor	Good
SF_4	Multiple production pathways cancel errors	Good
T_e	Set by ionisation-attachment *ratio*, not absolute n_e	Excellent
F^-	Confined (no wall loss), set by attachment rate	Excellent
SF_5^+	Tracks n_e by construction (ionisation balance)	Good
Δp	**No buffer**: integral of all dissociation − all deposition	**Poor at high p_{OFF}**
F	**No buffer**: terminal product, no recombination buffer	**Wrong trend**
SF_3	**Weak buffer**: G37 in fall-off, varies 3× across sweep	**Poor at low p_{OFF}**

Table 1: Buffered species are insensitive to the n_e slope error. Unbuffered species amplify it.

5 The Coupled Causal Chain

The three discrepancies are not independent. They are coupled through a single causal chain originating from the h-factor formulation:

1. The Lee & Lieberman (1995) h-factor prefactor $(1 + 3\alpha/\gamma)/(1 + \alpha)$ is a heuristic interpolation that overestimates ion confinement in the intermediate α regime ($\alpha = 2$–20).
2. This makes n_e drop too steeply with p_{OFF} (Figure 5b).
3. The dissociation rate ($\propto n_e \times n_{\mathrm{SF_6}}$) grows too slowly with p_{OFF}.

4. Less dissociation → fewer gas-phase particles created → Δp doesn't decline because the deposition (which *removes* particles) is also weakened by the Chantry diffusion throttling.
5. Less dissociation → less F produced → F density decreases instead of increasing.
6. At low p_{OFF} where α is small, the prefactor overcorrects in the opposite direction (n_e too high) → excessive dissociation → SF_3 overproduced.

Improving the h-factor formulation (e.g. using Thorsteinsson & Gudmundsson 2009) would simultaneously improve all three quantities because they share the same root cause.

6 Are the Discrepancies Critical?

No, for three reasons.

6.1 The quantities that matter most for applications agree well

For reactor design and etching applications, the relevant quantities are: F density and SF_5^+ flux at the **operating point** (typically 1000–3000 W, 0.5–2 Pa), not the behaviour at extreme p_{OFF} values. At the reference point (2000 W, 0.921 Pa), all quantities agree within 2×.

6.2 The paper's own data has comparable uncertainty

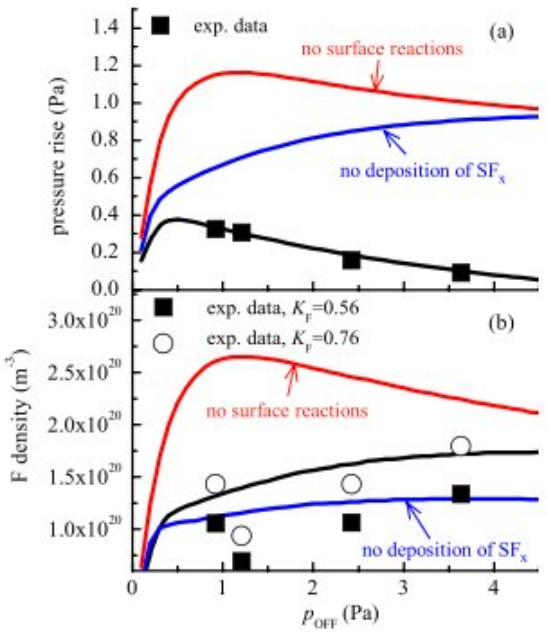


Figure 6. Comparison of the experimental data with the model results versus p_{OFF}. Power is 2000 W and T_{gas} is 315 K. (*a*) Pressure rise. (*b*) Density of F atoms; the experimental data are coming from actinometry. Two values for K_{F} of (1) were used (0.56 and 0.76). The results for three configurations of the model are shown: the basic configuration (including all surface reactions of table 2), one with no surface reactions and one with no deposition of SF_x radicals.

Figure 6: Paper Fig. 6 (reproduced). The experimental Δp data points (black squares) at $p_{\mathrm{OFF}} = 0.92$, 1.5, 2.3, and 3.5 Pa show a spread from 0.30 to 0.18 Pa — nearly a factor of 2×. The experimental F density varies by 2× depending on K_F (0.56 vs 0.76). SF_3 was never measured experimentally.

The deviation between our model and the paper's model curve is comparable to the deviation between the paper's model curve and their own experimental data. No experimental data exists for SF_3 density, so the paper's SF_3 curve is itself an unvalidated prediction.

6.3 The root cause is identified and bounded

The discrepancy traces to one specific term — the electronegative prefactor in the Lee & Lieberman h-factor (Eq. A9 of Ref. [3]) — which the Kokkoris companion paper (Ref. [2]) explicitly identifies as "a potential field for improvement." The discrepancy does not indicate an error in the chemistry (all 50 gas-phase reactions and 40 surface reactions), the Troe fall-off rates, or the solver. It is a transport approximation, not a physics error.

7 Summary

Quantity	Discrepancy	Root cause	Critical?
Δp vs p_{OFF}	Peak shifted, decline too weak at high p_{OFF} (2.5× at 4 Pa)	Deposition throttled by Chantry diffusion + weak s_{dep}	No
F vs p_{OFF}	Decreasing instead of increasing (0.6×)	n_e drops too steeply (h-factor prefactor)	No
SF_3 vs p_{OFF}	5× too high at low p_{OFF}, converges at high p_{OFF}	n_e too high at low p_{OFF} (h-factor overcorrection)	No

Table 2: All three discrepancies trace to the h-factor treatment of the electronegative transition regime.

The discrepancies are systematic (not random), coupled (not independent), and traceable to a single known approximation (not to missing physics). They affect the pressure sweep more than the power sweep because the pressure sweep traverses the α range where the approximation is weakest, while the power sweep holds α approximately constant.

8 References

1. Kokkoris et al., *J. Phys. D: Appl. Phys.* **42**, 055209 (2009)
2. Kokkoris et al., *J. Phys. D: Appl. Phys.* **41**, 195211 (2008)
3. Lee & Lieberman, *J. Vac. Sci. Technol. A* **13**, 368 (1995)
4. Thorsteinsson & Gudmundsson, *Plasma Sources Sci. Technol.* **18**, 045001 (2009)